

Mechanical Engineering – Pumps and Maintenance

Centrifugal Pump

A centrifugal pump converts mechanical energy from a rotating impeller into fluid energy. It is widely used for water circulation, cooling systems, and industrial fluid transfer.

Major Parts

Major components include the impeller, casing, shaft, bearings, mechanical seal, suction pipe, and discharge pipe.

Pump Operation

Fluid enters through the suction side and is accelerated by the rotating impeller. The casing converts part of the velocity energy into pressure energy and directs fluid toward the discharge outlet.

Common Faults

Common pump faults include overheating, abnormal vibration, low discharge pressure, leakage, unusual noise, and failure to prime.

Troubleshooting

For low discharge pressure, check the suction line, valve position, pump rotation, impeller condition, and blockage. For excessive vibration, inspect alignment, bearings, foundation, and rotating components.

Maintenance

Routine maintenance should include checking lubrication, seals, alignment, vibration, temperature, fasteners, and abnormal noise. Equipment should be isolated before mechanical inspection.

Safety

Disconnect and isolate the pump before maintenance. Relieve stored pressure and allow hot equipment to cool before inspection.